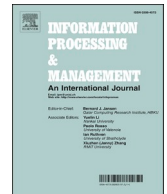




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BCMF: A bidirectional cross-modal fusion model for fake news detection

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ABSTRACT

In recent years, fake news detection has been a significant task attracting much attention. However, most current approaches utilize the features from a single modality, such as text or image, while the comprehensive fusion between features of different modalities has been ignored. To deal with the above problem, we propose a novel model named Bidirectional Cross-Modal Fusion (BCMF), which comprehensively integrates the textual and visual representations in a bidirectional manner. Specifically, the proposed model is decomposed into four submodules, i.e., the input embedding, the image2text fusion, the text2image fusion, and the prediction module. We conduct intensive experiments on four real-world datasets, i.e., Weibo, Twitter, Politi, and Gossip. The results show 2.2, 2.5, 4.9, and 3.1 percentage points of improvements in classification accuracy compared to the state-of-the-art methods on Weibo, Twitter, Politi, and Gossip, respectively. The experimental results suggest that the proposed model could better capture integrated information of different modalities and has high generalizability among different datasets. Further experiments suggest that the bidirectional fusions, the number of multi-attention heads, and the aggregating function could impact the performance of the cross-modal fake news detection. The research sheds light on the role of bidirectional cross-modal fusion in leveraging multi-modal information to improve the effect of fake news detection.

1. Introduction

The increasing number of Internet users and the opening trend of social platforms has led to the proliferation of fake news, i.e., news with fabricated misinformation, which is harmful to cyberspace and has adverse effects in the real world (Zhang and Ghorbani, 2020). As a result, fake news detection has been a significant task attracting attention. To facilitate fake news detection, researchers have proposed many approaches, which can be basically summarized into two categories, i.e., traditional machine-learning-based approaches (Ma, Gao, Wei, Lu, & Wong, 2015) and deep-learning-based approaches (Girgis and Gadallah, 2018). The first category mainly feeds the manual-defined features into a predefined classifier such as decision tree (Hakak et al., 2021). As the performance of these methods is highly dependent on the definition of hand-crafted features, it leads to low generalizability among different domains. The second one usually leverages deep neural networks to extract features automatically. For instance, Yu et al. (2017) employed a convolutional neural network (CNN) to obtain latent features concerning the text. Although great improvements have been achieved to detect fake news, most of above approaches utilize the features from a single modality.

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In recent years, as deep neural networks have yielded great success in natural language processing, researchers attempted to conduct multi-modal analysis by leveraging deep learning to obtain representative features of different modalities. Wang et al. (2018) proposed the EANN model, which leverages TextCNN (Kim 2014) to extract textual features and VGG-19 (Simonyan and Zisserman, 2015) to extract visual features, and the concatenated features are used to calculate the loss of the real-fake news classification and the event-type classification. Similarly, the MVAE model (Khattar et al., 2019) used a variational autoencoder to capture the latent features of multi-modalities, and the multi-modal features are concatenated and fed into an end-to-end network coupled with a classifier to predict whether the news is fake or real. The above experiments suggest that the integration of features of different modalities could achieve better performance than the single-modality analysis.

Although much progress has been made in multi-modal fake news detection, there still exist the following problems. First, although some existing models have attempted to leverage the multi-modal representations to enhance performance, they usually employ the last layer of pre-trained models such as BERT (Devlin, Chang, Lee, and Toutanova, 2019) to achieve textual representation. In contrast, the multi-modal contextual representation information has been ignored. Second, although some research has attempted to combine different modalities' representations, the aggregation between visual and textual features is still insufficient. It needs to take full advantage of the complementary information and the similarities between the information of different modalities. Third, to the best of our knowledge, most of the current research investigates the performance of fake news detection on one type of data, i.e., domain-independent or domain-specific datasets. However, the investigation of the generalization ability among domain-independent and domain-specific datasets is not sufficient.

To solve the above problems, a novel bidirectional cross-modal fusion model is proposed in this study to strengthen the integration of information from different modalities. Specifically, the contextualized pre-trained visual model, i.e., DeiT (Touvron et al., 2021), is used to replace the traditional visual model. Further, two bidirectional fusion modules, i.e., the image2text fusion and the text2image fusion, are proposed to aggregate and enhance the multi-modal representations. Specifically, the main contributions of this paper are as follows.

(1) We propose a novel model, namely Bidirectional Cross-Modal Fusion (BCMF), for fake news detection. To the best of our knowledge, this is one of the earliest attempts to leverage both contextualized visual embeddings and bi-directional fusions to facilitate fake news detection.

(2) We propose a bi-directional cross-modal aggregation mechanism to deeply fuse the visual and textual information, which could take full advantage of the complementary information between different modalities.

(3) Extensive experiments are conducted on four real-world datasets, i.e., Weibo, Twitter, Politi, and Gossip. The experimental results demonstrate the effectiveness and generalizability of our proposed approach on both domain-independent and domain-specific datasets.

The rest of the paper is organized as follows. In Section 2, the related work is reviewed. Our proposed model BCMF is presented in Section 3. Then extensive experiments are conducted, and the results are analyzed in Section 4. Finally, the findings are concluded in Section 5.

2. Related work

In most of the current researches, fake news detection is formulated as a binary classification problem. According to the information utilized, fake news detection could be categorized into single-modal and cross-modal analyses.

2.1. Fake news detection based on single-modality

In single-modal analysis, researchers of fake news detection mainly leverage information of single-modality to capture latent features for the determination of fake news.

Traditional studies of fake news detection focus on extracting statistical or semantic information from the textual content of the posts. For instance, Ruchansky, Seo, and Liu (2017) used a long short-term memory network (LSTM) to extract textual features for articles and users. Ma et al. (2016) utilized a recurrent neural network (RNN) to capture salient features from the relevant posts. Besides RNN, researchers leveraged the convolutional neural network (CNN) (Yu et al., 2017) or CNN with margin loss (Goldani, Safabakhsh, and Momtazi, 2021) to obtain features. Emotional features in the text have also been considered for fake news detection (Ajao, Bhowmik, & Zargari, 2019; Cui, Wang, & Lee, 2019; Kumari, Ashok, Ghosal, & Ekbal, 2021b; Kumari et al., 2021a; Zhang et al., 2021). In addition to sentimental analysis, researchers have also attempted to use textual style (Potthast, Kiesel, Reinartz, Bevendorff, and Stein, 2018) and linguistic features (Choudhary & Arora, 2021; Chu, Xie, & Wang, 2021; Li, Fan, Yuan, and Zhang, 2021) to detect fake news.

Compared with the text-based approaches, minor literature uses single visual features to detect fake news. Previous work (Jin, Cao, Zhang, Zhou, and Tian, 2017) has found that images concerning fake and real news contain different image distributions. Thus, the

statistical features of single-modality can be leveraged to detect fake news. Qi, Cao, Yang, Guo, and Li (2019) investigated the feasibility of only using visual features to detect fake news. Two neural networks are utilized to extract the complex patterns and semantic features from the frequency domain and the pixel domain, respectively. However, as the visual features are not integrated with the textual contents, the above approaches have not achieved state-of-the-art performances.

2.2. Fake news detection based on multi-modality

In multi-modal analysis, researchers mainly adopt both visual and textual features to differentiate fake news.

The early studies of multi-modal fake news detection attempted to integrate features of different modalities in the way of concatenation. For instance, Jin et al. (2017) proposed the att-RNN model, which leverages a recurrent neural network augmented with an attention mechanism to fuse features from different modalities. Specifically, the text and social context information are combined and fed into a long short-term memory network. Then the output is dot-multiplied with the visual features extracted by VGG-19 to obtain the integrated features to produce a reliable prediction. Wang et al. (2018) proposed the EANN model, which leverages TextCNN to extract textual features and VGG-19 to extract visual features. The concatenated features are input into a fake news detector to predict the authenticity of the news. Meanwhile, the features are input into an event discriminator to classify the event types. Similarly, the MVAE model (Khattar et al., 2019) used a variational autoencoder to capture the latent features of multi-modalities. The multi-modal features are concatenated and fed into an end-to-end network coupled with a classifier to predict whether the news is fake or real. The above research suggested that the integration of features of different modalities could achieve better performance than the single-modality analysis. However, as the aggregation between visual and textual features is restricted to concatenation, there is still insufficient to take full advantage of information from different modalities.

To address the above issue, researchers attempted to examine the correlation between different modalities in a deep manner and improve the fusion mode by introducing a more advanced aggregating mode. For instance, Zhou et al. (2020) assume that fake news could be identified by quantitatively evaluating the semantic distance between the visual and corresponding textual information. Based on this, the irrelevance between different modalities is integrated into the loss function to enhance the performance of the predictor. Similarly, the AMFB model (Kumari and Ekbal, 2021) used a multi-modal factorized bilinear pooling to maximize the correlations between the features of different modalities and achieve their alignments. HMCAN (Qian et al., 2021) used two contextual Transformers with multi-head attention and layer-norms to capture the multi-modal contextual information. CARMN (Song, Ning, Zhang, and Wu, 2021) utilized two cross-modal attention residual networks to capture the bidirectional attention between different modalities. Their experimental results suggest that introducing advanced integration between different modalities could improve the performance of the cross-modal analysis.

With the contextualized pre-trained language models, such as BERT (Devlin, Chang, Lee, and Toutanova, 2019), RoBERTa (Liu et al., 2019), and XLNet (Yang et al., 2019), having yielded great success in natural language processing, researchers realize that improving the textual representations play an essential role in differentiating fake news. Recently, multi-modal analysis with contextualized pre-trained embeddings has received increasing attention. For instance, the Spotfake model (Singhal et al. 2019) used the contextualized pre-trained embedding, i.e., BERT, to enhance the textual representation. The textual embeddings are concatenated with the image embeddings, i.e., VGG19, then fed into the neural networks for prediction. Similarly, the Spotfake+ model (Singhal et al. 2020) utilized XLNet to obtain pre-trained textual embeddings. Samadi, Mousavian, and Momtazi (2021) employed RoBERTa (Liu et al., 2019) and other contextualized models to extract sentence features. The experimental results suggest that the Transformer-based textual models could improve the performance of the cross-modal analysis. However, from the visual perspective,

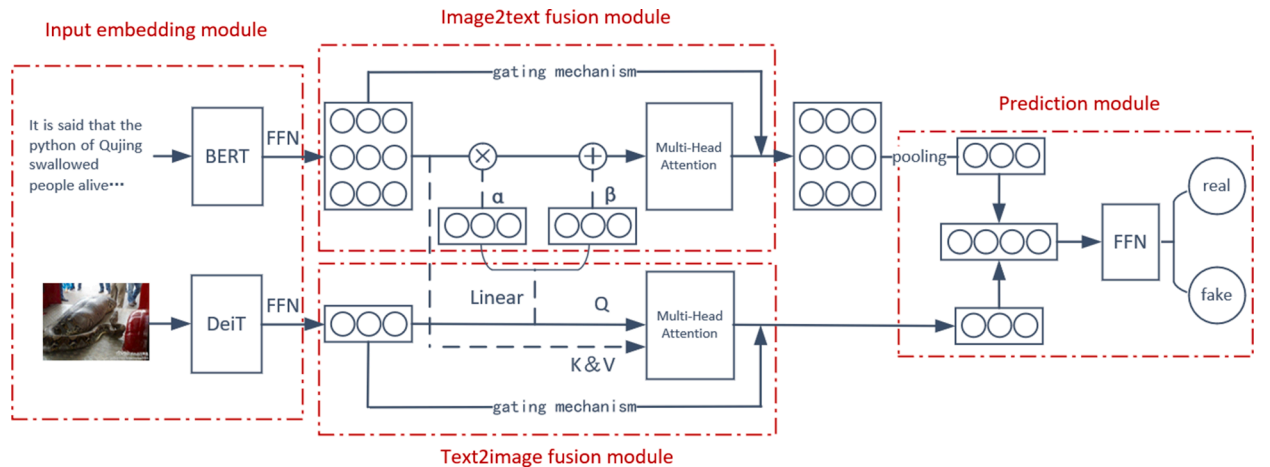


Fig. 1. The architecture of the proposed bi-directional cross-model fusion model for fake news detection

the convolutional architecture such as VGG-19 remains dominant in cross-modal fake news detection. Although recently several researchers have attempted to transplant the Trans former ingredients to vision tasks such as image classification (Dosovitskiy et al., 2021; Touvron et al., 2021), few literatures have investigated the role of the contextualized visual models such as ViT (Dosovitskiy et al., 2021) and DeiT (Touvron et al., 2021) in cross-modal fake news detection.

In summary, although recent research has significantly improved multi-modal fake news detection, they are still insufficient in leveraging the multi-modal contextual and complementary information. The deep fusion between the features of different modalities is largely unsolved. To address the above issues, we propose a bidirectional cross-modal fusion (BCMF) model by leveraging multi-modal contextual information to jointly model the bidirectional cross-modal fusion.

3. Research framework

3.1. Research questions

Fake news detection can be formulated as a binary classification task. Consider a post to be judged is denoted as $P = (T, I)$, where $T = \{w_1, w_2, \dots, w_n\}$ indicates the textual content composed of n words, and I denotes the image attached to the post. The model will learn a fake news detection function, i.e., $F(T, I) \rightarrow (\hat{y})$, where $\hat{y}=0$ and $\hat{y} = 1$ represent the news to be authentic and fake, respectively. Specifically, this study mainly investigates the following research questions.

RQ1: Do the fake news detection models based on multi-modalities have a significantly better performance than traditional algorithms of single-modality?

RQ2: Among the methods of multi-modalities, is the performance of the BCMF model better than that of state-of-the-art methods?

RQ3: In terms of BCMF, how does the model architecture influence the performance? Specifically, we investigate: (1) can bi-directional fusions, i.e., the image2text fusion and the text2image fusion, effectively improve the performance? (2) does the number of multi-attention heads influence the model performance? and (3) does the aggregating function influence the performance?

3.2. Model framework

As shown in Fig. 1, the proposed Bidirectional Cross-Modal Fusion (BCMF) model is divided into four modules, i.e., the input embedding, the image2text fusion, the text2image fusion, and the prediction modules.

3.2.1. The input embedding module

As mentioned in the research questions, the input of the BCMF model is a multi-modal post $P = (T, I)$, where T indicates the textual content composed of n words, and I denotes the image attached to the post.

To capture the contextual textual embedding, we utilize Bidirectional Encoder Representations from Transformers (BERT) (Devlin, Chang, Lee, and Toutanova, 2019), which has been proven to be the most effective in the natural language processing related task, as the core module for text encoding. Consider the textual content of the post is represented as $T = \{w_1, w_2, \dots, w_n\}$, where n_t is the length of the text. BERT is utilized to obtain the initial word embedding $R_{\text{BERT}} \in \mathbb{R}^{n_t \times d_t}$, where d_t is the dimension of the textual embedding.

To capture the contextual visual embedding, we utilize Data-efficient Transformers (DeiT) (Touvron et al., 2021), which has been proved to be effective in the computer vision related task, as the core module for visual encoding. Consider the visual content of the post is represented as $I \in \mathbb{R}^{H \times W \times C}$, where H , W , and C indicate the image height, the image width, and the number of channels, respectively. DeiT is utilized to obtain the initial image embedding $R_{\text{DeiT}} \in \mathbb{R}^{n_v \times d_v}$, where n_v indicates the number of images and d_v represents the dimension of the visual embedding. Following previous work (Song, Ning, Zhang, and Wu, 2021), we set $n_v=1$ in this paper.

The above process is defined as shown in Eq. (1) and Eq. (2).

$$R_t = \text{FFN}(\text{BERT}(T)) \quad (1)$$

$$R_v = \text{FFN}(\text{DeiT}(I)) \quad (2)$$

where R_t and R_v indicate the initialize embedding of the textual and visual content, respectively, and FFN represent the feed-forward neural network.

3.2.2. The image2text fusion module

Given a pair of text and image in a post, the task of the image2text fusion module is to fuse the relevant and complementary visual information into the textual embedding. It consists of three components, i.e., the adaptation component, the self-attention component, and the gate component.

The adaptation component filters the features from the visual modality into the textual modality. Inspired by the work of [Wehrmann et al. \(2020\)](#), two linear projections are used upon the visual representation features R_v to obtain a scale vector and a shift vector, which are denoted by α ($\alpha \in \mathbb{R}^{n_v \times d}$) and β ($\beta \in \mathbb{R}^{n_v \times d}$), respectively, as shown in [Eq. \(3\)](#) and [\(4\)](#). Upon α and β , the visual features R_v is adapted to the textual features R_t by adding β to the point-wise multiplication between each row of R_t and α , as shown in [Eq. \(5\)](#).

$$\alpha = R_v \times W_\alpha + b_\alpha \quad (3)$$

$$\beta = R_v \times W_\beta + b_\beta \quad (4)$$

$$R_{v \rightarrow t} = R_t * \alpha + \beta \quad (5)$$

where $W_\alpha \in \mathbb{R}^{d \times d}$ and $W_\beta \in \mathbb{R}^{d \times d}$ are the weights of the neural network, $b_\alpha \in \mathbb{R}^d$ and $b_\beta \in \mathbb{R}^d$ are the biases of the neural network, $*$ indicates the point-wise multiplication between each row of R_t and α , which generates the visual-adapted textual features, i.e., $R_{v \rightarrow t}$.

The self-attention component aims to calculate the amount of attention to be paid to different words of a sentence. Upon the visual-adapted textual features $R_{v \rightarrow t}$, three matrices, $Q \in \mathbb{R}^{n_t \times d_k}$, $K \in \mathbb{R}^{n_t \times d_k}$, and $V \in \mathbb{R}^{n_t \times d_v}$ are generated linearly through [Eq. \(6\)](#), [\(7\)](#), and [\(8\)](#), respectively.

$$Q = R_{v \rightarrow t} \times W_Q \quad (6)$$

$$K = R_{v \rightarrow t} \times W_K \quad (7)$$

$$V = R_{v \rightarrow t} \times W_V \quad (8)$$

where $W_Q \in \mathbb{R}^{d \times d_k}$, $W_K \in \mathbb{R}^{d \times d_k}$, and $W_V \in \mathbb{R}^{d \times d_v}$ are the weights of the neural network, and $d_k = d_v = d$.

Using the generated query (Q), key (K) and value (V), the attention score between different words indicating the relative correlation will be obtained. As it is found by prior research ([Vaswani et al., 2017](#); [Song et al., 2021](#)) that the multi-head attention can make more effective use of the contextualized information than the single-head attention. On this account, the original Q, K, and V are divided into H subspaces, and the h -th head are denoted by $Q_h \in \mathbb{R}^{n_t \times \frac{d_k}{H}}$, $K_h \in \mathbb{R}^{n_t \times \frac{d_k}{H}}$, and $V_h \in \mathbb{R}^{n_t \times \frac{d_v}{H}}$, respectively. The features obtained with respect to the single-head attention, i.e., $\widetilde{R}_h$, is shown in [Eq. \(9\)](#). The outputs of all the heads are concatenated to form the self-attentional and visual-adapted textual features, i.e., $R'_{v \rightarrow t}$, as shown in [Eq. \(10\)](#).

$$\widetilde{R}_h = \text{softmax} \left(\frac{Q_h \times K_h^T}{\sqrt{d_k/H}} \right) \times V_h \quad (9)$$

$$R'_{v \rightarrow t} = \text{Concat}([\widetilde{R}_{h1}, \widetilde{R}_{h2}, \dots, \widetilde{R}_{hH}]) \quad (10)$$

where H is the number of the multi-head attentions, and Concat() is the concatenation function of vectors.

The gating component is used to learn a coefficient to determine how much the visual representation should be integrated into the normal textual representation. Specifically, the concatenation of the original text feature R_t and the visual-adapted textual features $R'_{v \rightarrow t}$ is fed into two feedforward neural networks with different activation functions, i.e., Sigmoid and Tanh. Sigmoid is used as the first activation function to obtain a gate, i.e., f_t , which indicates the proportion of the original textual representation to be kept. Tanh is used as the second activation function to obtain the updated features, i.e., $\widetilde{C}_t$. The calculation is shown in [Eq. \(11\)](#), [\(12\)](#) and [\(13\)](#).

$$f_t = \text{sigmoid}(W_{f1} \times [R_t, R'_{v \rightarrow t}] + b_{f1}) \quad (11)$$

$$\widetilde{C}_t = \text{tanh}(W_{C1} \times [R_t, R'_{v \rightarrow t}] + b_{C1}) \quad (12)$$

$$R'_t = f_t * R_t + (1 - f_t) * \widetilde{C}_t \quad (13)$$

where W_{f1} and W_{C1} are weights of the networks, b_{f1} and b_{C1} are bias of the networks, and R'_t is the fused textual features.

3.2.3. The text2image fusion module

Given a pair of text and image, the task of the text2image fusion module is to fuse the relevant and complementary textual information into the image embedding. It consists of two components, i.e., the text-to-image attention component and the gate component.

The text-to-image attention component measures the dependence between each word and the image. Upon the visual features R_v

and the textual features R_t , three matrices, i.e., $Q = R_v \times W_Q$, $K = R_t \times W_K$, and $V = R_t \times W_V$, are generated, respectively. Using the generated query (Q), key (K) and value (V), the attention score between each word and image indicating the relative dependence will be obtained. Using Eq. (9) and (10), we achieve the textual-adapted visual features $R'_{t \rightarrow v}$.

The gate component is used to learn a coefficient to determine how much the textual representation should be integrated into the normal visual representation. It combines the original visual feature R_v and the textual-adapted features $R'_{t \rightarrow v}$. The concatenation is fed into two feedforward neural networks with different activation functions to obtain a gate, i.e., f_v , which indicates the proportion of the original visual representation to be kept, and the updated features, i.e., $\widetilde{C}_v$. The calculation is shown in Eq. (14), (15), and (16).

$$f_v = \text{sigmoid}(W_{f2} \times [R_v, R'_{t \rightarrow v}] + b_{f2}) \quad (14)$$

$$\widetilde{C}_v = \tanh(W_{C2} \times [R_v, R'_{t \rightarrow v}] + b_{C2}) \quad (15)$$

$$R'_v = f_v * R_v + (1 - f_v) * \widetilde{C}_v \quad (16)$$

where W_{f2} and W_{C2} are weights of the networks, b_{f2} and b_{C2} are biases of the networks, and R'_v is the fused visual features.

3.2.4. The prediction module

The prediction module is to leverage the features extracted from the model to classify the authenticity of news according to its advanced semantics. It is suggested by prior research (Zhou, Wu, and Zafarani, 2020) that the similarity of information between different modalities could be used to enhance the effect of fake news detection. Inspired by this, we follow the work of Reimers and Gurevych (2019) by concatenating $\text{Maxpool}(R'_t)$, R'_v , and $|\text{Maxpool}(R'_t) - R'_v|$. The calculation is defined as:

$$\overline{R}_f = \text{Concat}([\text{Maxpool}(R'_t), R'_v, |\text{Maxpool}(R'_t) - R'_v|]) \quad (17)$$

where $\overline{R}_f$ is the fused features, $\text{Maxpool}()$ is the max pooling function, and $\text{Concat}()$ is the concatenation function of vectors.

The fused features $\overline{R}_f$ are finally passed through a feed-forward neural network (FFN) and then fed into a Softmax function to calculate the probability of the sample being true or fake. The cross entropy is adopted as the loss function. The prediction and loss calculation are shown in Eq. (18) and (19), respectively.

$$\hat{y} = \text{Softmax}(\text{FFN}(\overline{R}_f)) \quad (18)$$

$$L(\theta) = -y \log(\hat{y}_1) - (1 - y) \log(\hat{y}_0) \quad (19)$$

where $\hat{y} = [\hat{y}_0, \hat{y}_1]$ ($\hat{y}_0$ denotes a particular news to be true, and $\hat{y}_1$ represents it to be fake), FFN is a feedforward neural network, and θ indicates the model parameters.

4. Experiments

4.1. The datasets

To verify the effect of the proposed model, we conduct empirical analysis on two domain-independent datasets, i.e., Weibo and Twitter, and two domain-specific datasets, i.e., Politi and Gossip. The statistical information of the datasets is shown in Table 1.

1) Weibo dataset. It is a Chinese dataset firstly presented by Jin et al. (2017). The real news came from authoritative Chinese news sources such as Xinhua. In contrast, the fake news came from posts between May 2012 and January 2016, verified as counterfeit by the official Weibo rumor-debunking system. Following the previous work of Wang et al. (2018), the low-quality images have been removed from the dataset to ensure the overall quality. In addition, posts containing only text are removed to ensure that each post in the data set has textual information and image associated with it.

2) Twitter dataset. It is an English dataset presented by Boididou et al. (2016) as a part of the Verifying Multimedia Use task. As the original information were collected by retrieving different events, there are no repetitive contents of events between the training and test set. We only keep the tweets that consist of both textual and visual information.

3) Politi dataset. It is a domain-specific dataset presented by Shu et al. (2020). The Politi dataset comes from PolitiFact, a website where political news posts are reviewed by journalists and field experts and provided fact-checking assessments to label posts as authentic or fake. Following the previous work of Singhal et al. (2020), the dataset was preprocessed by removing posts without images.

4) Gossip dataset. It is another domain-specific dataset presented by Shu et al. (2020). The Gossip dataset comes from two websites, i.e., GossipCop and E!Online. GossipCop is a website that fact-checks entertainment news gathered by various media outlets, rating each post on a credibility scale of 0 to 10. The news on GossipCop is unbalanced, as it is found that 90 percent of the posts had a score of less than 5. To resolve this issue, the E!Online website, which is a trusted entertainment news posting site that aggregates real entertainment news, is chosen as a supplementary source for real news. In contrast, the news with a score under 5 on GossipCop is taken as a fake sample.

Table 1
The statistics of datasets.

Datasets	Weibo	Twitter	Politi	Gossip
#Real News	4,779	6,026	642	16,817
#Fake News	4,749	7,898	432	5,323
#Images	9,528	514	783	18,417

Table 2
The configuration of the hyperparameters

hyperparameters	value
optimizer	Adam
epoch	20
batch size	16
dropout	0.2
learning rate	1e-5
d_t, d_v, d_k	768
FFN hidden dim	256

4.2. Implementation details

In this study, we implemented the model under the CUDA 11.0 acceleration with a GeForce RTX 2080Ti (11GB of memory) GPU. The coding environment was the Pytorch framework. In addition, Table 2 shows the configuration of the hyperparameters.

Following previous work (Song, Ning, Zhang, and Wu, 2021) on cross-modal fake news detection, we use Adam (Kingma and Ba, 2014) as the optimizer. The data are trained for 20 epochs, and 16 samples are sent into each batch. In order to avoid over-fitting problems, the dropout rate of the neural network is set to 0.2, and the learning rate is set to 1e-5. The embedding dimension d_k in the multi-head attention layer is set to 768. In addition, in the visual representation module, the hidden dimension of the feedforward neural network (FFN) is 256. To ensure the training and testing set does not contain the same event, we comply with the previous work (Jin et al., 2017; Singhal et al., 2019; Qian et al., 2021) by dividing the training and testing sets with an approximate proportion of 8:2. Specifically, the sizes of training (testing) set on the Weibo, Twitter, Politi, and Gossip are 7,532 (1,996), 12,933 (991), 381 (104), and 10,010 (2,830), respectively.

The run time cost varies concerning different datasets. In this study, we use BERT-Base-Chinese¹ (102M) for the Weibo dataset and BERT-Base-Uncased² (110M) for the Twitter, Politi, and Gossip datasets. Based on this, for the Chinese dataset, the number of parameters of the BCMF model is 110M; For the English dataset, the size of the BCMF model is 118M. More specifically, it takes 0.120, 0.107, 0.126, and 0.126 seconds to predict the authenticity of 100 news on Weibo, Twitter, Politi, and Gossip, respectively.

4.3. Baseline methods

To verify the effectiveness of the proposed model, we compare the experimental results on the four real-world datasets, i.e., Weibo, Twitter, Politi, and Gossip, with the following state-of-the-art methods.

Bi-LSTM (Khatter et al. 2019) and CNN (Singhal et al., 2020): These are models of single-modality that utilize only textual representations. Specifically, Bi-LSTM and CNN employ a bi-directional long short term memory network and a convolutional neural network to extract textual features, respectively.

VGG-19 (Khatter et al. 2019): It is model of single-modality that utilizes only visual representations. Specifically, the attached images for a post are fed into a VGG-19 network to extract visual features.

att-RNN and att-RNN* (Jin et al., 2017): The att-RNN model leverages a recurrent neural network augmented with an attention mechanism to fuse features from different modalities. Specifically, the text and social context information are concatenated, and an LSTM network is used to further extract features. The output is then dot-multiplied with the visual features extracted by VGG-19 to obtain the integrated features to produce a reliable prediction. The att-RNN* model is a variation of att-RNN, which removes the social context information from the original model to make a fair comparison among different models for cross-modal fake news detection.

EANN (Wang et al., 2018) and EANN* (Khatter et al., 2019): The EANN model leverages TextCNN to extract textual features and VGG-19 to extract visual features. The concatenated features are used to calculate the loss of the real-fake news classification and the event-type classification. In addition, the model introduces an event discriminator as an adversarial network to learn the shared features of fake news among different events and remove those event-dependent features. The EANN* model is a variation of EANN, which removes the event discriminator to make a fair comparison.

MVAE (Khatter et al. 2019): The model uses a variational autoencoder to capture the latent features of multi-modalities.

¹ <https://huggingface.co/bert-base-chinese>.

² <https://huggingface.co/bert-base-uncased>.

Specifically, BiLSTM is used to extract text features, and VGG-19 is used to extract image features. The multi-modal features are concatenated and fed into the end-to-end network coupled with a classifier to predict whether the news is fake or real.

SAFE (Zhou et al. 2020): The model assumes that fake news could be identified by quantitatively evaluating the semantic distance between the visual information and the corresponding textual information. The irrelevance between different modalities is integrated into the loss function to enhance the predictor's performance.

Spotfake (Singhal et al., 2019) and Spotfake+ (Singhal et al., 2020): The Spotfake model uses the contextualized pre-trained embedding, i.e., BERT, to enhance the textual representation. The textual embeddings are concatenated with the image embeddings, i.e., VGG-19, then fed into the neural networks for prediction. The Spotfake+ model is a variation of Spotfake, which replace the text feature extraction module with XLNet (Yang et al., 2019).

Singh and Sharma (2021): The model leverages EfficientNetB0 to learn the features of images processed by an error level analysis (ELA), which has been used as a forensic approach to capture the difference between images in the compressed format. In addition, the contextualized pre-trained embedding, i.e., RoBERTa (Liu et al., 2019), is used to extract sentence features.

AMFB (Kumari and Ekbal, 2021). The model employs a stacked bi-directional LSTM and a CNN-RNN coupled network to extract the features for texts and images, respectively. Based on this, a multi-modal factorized bilinear pooling is utilized to maximize the correlations between the features of different modalities and achieve their alignments.

HMCAN (Qian et al., 2021): The model employs BERT and ResNet50 to generate the embedding vectors for words and images, respectively. Based on this, two contextual Transformers with multi-head attention and layer-norms are utilized to capture the multi-modal context information.

CARMN (Song, Ning, Zhang, and Wu, 2021): The model utilizes word2vec and VGG-19 to extract initial text and image embeddings, respectively. Based on this, two cross-modal attention residual networks are employed to capture the bidirectional attention, and a multi-channel CNN is employed to extract textual features further.

4.4. Comparison of experimental results

Table 3 and Table 4 demonstrate the experimental results of the proposed BCMF model and all baseline models on the domain-independent datasets, i.e., Weibo and Twitter, and the domain-specific datasets, i.e., Politi and Gossip. We have the following observations.

On the two domain-independent datasets, it can be seen from Table 3 that the proposed BCMF model has achieved better results than the baseline approaches. Specifically, on the Weibo dataset, our model has achieved an increment of 2.2 points in Acc, compared with the HMCAN model, which has the best performance among the baseline approaches. Similarly, on the Twitter dataset, our model has achieved an increment of 2.5 points in Acc, compared with the Spotfake+ model, which has the best performance among the baseline approaches. The experimental results suggest that the proposed model could achieve stable performance on Chinese and English datasets. In addition, compared with the models leveraging the single-modal representation, i.e., Bi-LSTM and VGG-19, the multi-modal models achieve significantly better performance concerning all indicators. It suggests that the integration of textual and visual representations can enhance the performance of fake news detection.

Table 3

The results of comparison among different models on the Weibo and Twitter datasets. The att-RNN* model is a variation of att-RNN, which removes the social context information from the original model to make a fair comparison. The EANN* model is a variation of EANN, which removes the event discriminator to make a fair comparison. “-” indicates that the results of this indicator are not provided in the original baseline paper.

Datasets	Models	Acc	Fake news			Real news		
			Pre	Recall	F1	Pre	Recall	F1
Weibo	Bi-LSTM	0.643	0.662	0.578	0.617	0.609	0.685	0.647
	VGG-19	0.608	0.610	0.605	0.607	0.607	0.611	0.609
	att-RNN	0.788	0.862	0.686	0.764	0.738	0.890	0.807
	att-RNN*	0.772	0.854	0.656	0.742	0.720	0.889	0.795
	EANN	0.827	-	-	-	-	-	-
	EANN*	0.782	0.827	0.697	0.756	0.752	0.863	0.804
	MVAE	0.824	0.854	0.769	0.809	0.802	0.875	0.837
	SAFE	0.763	0.833	0.659	0.736	0.717	0.868	0.785
	Singh&Sharma	0.812	0.851	0.784	0.816	0.744	0.826	0.782
	AMFB	0.832	0.820	0.860	0.840	0.850	0.810	0.830
	HMCAN	0.885	0.920	0.845	0.881	0.856	0.926	0.890
	BCMF (ours)	0.907	0.927	0.889	0.908	0.887	0.925	0.906
Twitter	Bi-LSTM	0.526	0.586	0.553	0.569	0.469	0.526	0.496
	VGG-19	0.596	0.695	0.518	0.593	0.524	0.700	0.599
	att-RNN	0.682	0.780	0.615	0.689	0.603	0.770	0.676
	att-RNN*	0.664	0.749	0.615	0.676	0.589	0.728	0.651
	EANN	0.715	-	-	-	-	-	-
	EANN*	0.648	0.810	0.498	0.617	0.584	0.759	0.660
	MVAE	0.745	0.801	0.719	0.758	0.689	0.777	0.730
	SpotFake+	0.790	0.793	0.827	0.810	0.786	0.747	0.766
	CARMN	0.741	0.854	0.619	0.718	0.670	0.880	0.760
	BCMF (ours)	0.815	0.854	0.805	0.829	0.772	0.827	0.799

Table 4

The results of comparison among different models on the Politi and Gossip datasets (Acc).

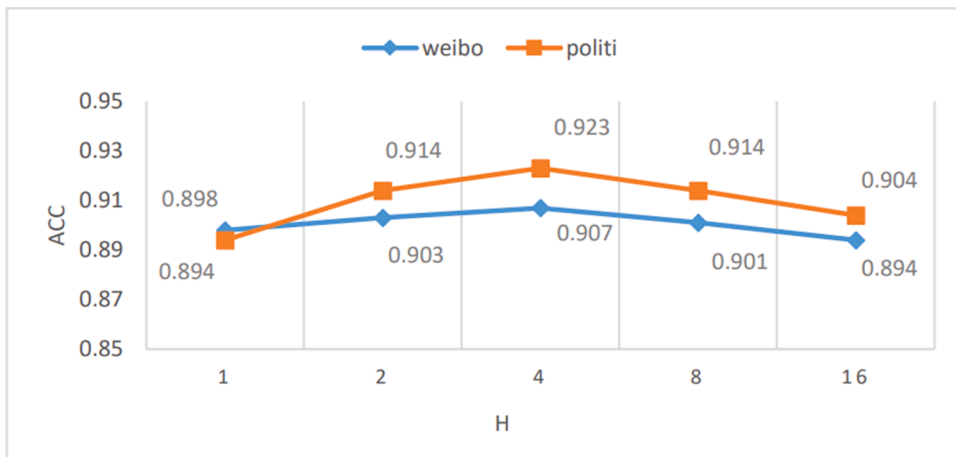
Models	Politi	Gossip
CNN	0.629	0.723
VGG-19	0.654	0.800
att-RNN*	0.769	0.743
EANN	0.740	0.860
MVAE	0.673	0.775
SAFE	0.874	0.838
SpotFake	0.721	0.807
SpotFake+	0.846	0.856
BCMF (ours)	0.923	0.891

Table 5

The ablation analysis of the BCMF model. “-image2text” denotes that the image2text fusion module is removed from the BCMF model; “-text2image” indicates that the text2image fusion module is removed from the BCMF model; “-both” represents that the model concatenates the contextualized pre-trained image and text features to predict the authenticity of the news.

Datasets	Model	Acc	Fake news Pre	Recall	F1	Real news Pre	Recall	F1
Weibo	-image2text	0.898	0.900	0.902	0.901	0.895	0.893	0.894
	-text2image	0.897	0.932	0.864	0.896	0.865	0.932	0.898
	-both	0.887	0.898	0.882	0.890	0.877	0.893	0.885
	BCMF	0.907	0.927	0.889	0.908	0.887	0.925	0.906
Politi	-image2text	0.904	0.771	0.931	0.844	0.971	0.893	0.931
	-text2image	0.894	0.800	0.828	0.814	0.932	0.920	0.926
	-both	0.885	0.815	0.759	0.786	0.909	0.933	0.921
	BCMF	0.923	0.862	0.862	0.862	0.947	0.947	0.947

On the two domain-specific datasets, we have similar observations. Specifically, on the Politi dataset, our model has achieved an increment of 4.9 points in Acc, compared with the SAFE model, which has the best performance among the baseline approaches. Similarly, on the Gossip dataset, our model has achieved an increment of 3.1 points in Acc, compared with the EANN model, which has the best performance among the baseline approaches. The experimental results further prove the advantages of the proposed BCMF model upon the baselines. In addition, compared with the models leveraging the single-modal representation, i.e., CNN and VGG-19, the multi-modal models achieve significantly better performance concerning all indicators. The experimental results further suggest that integrating textual and visual representations can enhance the performance of fake news detection.

**Fig. 2.** The influence of the number of multiple-attention heads on the performance of the BCMF model

4.5. Analysis of the ablation study

To investigate the influence of different fusion modules on the performance, ablation experiments are conducted on the domain-independent Weibo dataset and the domain-specific Politi dataset. The experimental results are shown in Table 5.

As can be seen from the experimental results in Table 5, the complete BCMF model achieved the best performance on both Chinese and English data sets. After the image2text fusion module is removed, the accuracy of the BCMF model on Weibo and Politi data sets decreases by 0.9 and 1.9 points, respectively. After removing the text2image fusion module, the accuracy of the Weibo and Politi datasets decreases by 1.0 and 2.9 points, respectively. In addition, the model accuracy drops by 2.0 and 3.8 points on the Weibo and Politi data sets, respectively, when the bidirectional fusion modules are removed. The experimental results suggest that both image2text and text2image fusion effectively improve the performance of fake news identification. Additionally, we observed that the effect decreases more obviously after removing the text2image module than the image2text module. It implies that the direction of the fusions influences the improving effect. Specifically, fusing the textual modality into the visual modality achieves a more remarkable improvement than the fusion of the inverse direction. This difference may be attributed to the reason that the textual and visual information did not play an equal role in the cross-modal fake news detection.

4.6. Analysis of multiple-attention heads

To investigate the influence of the number of multi-attention heads on the performance of the proposed model, we conduct a comparative experiment on the Weibo and Politi datasets. The influence of the number of multi-attention heads on the model is shown in Fig. 2.

As can be seen from Fig. 2, on both Weibo and Politi datasets, the Acc values of BCMF increase with the increasing number of multi-attention heads at the beginning. When multi-attention heads increase to 4, the model achieves optimal performance on both Weibo and Politi datasets, with the accuracy reaching 90.7% and 92.3%, respectively. Since then, the Acc values on both data sets have declined. The consistency of the experimental results on the two data sets shows the existence of the optimum number of multi-attention heads.

4.7. Analysis of the feature aggregating functions

As BCMF adopts a gate mechanism in both the image2text fusion module and the text2image fusion module, we further conduct an extended experiment on the Weibo and Politi datasets to investigate the influence of aggregating functions on the model performance. The experimental results are shown in Table 6.

As can be seen from Table 6, the model with the gate mechanism proposed in this paper achieves the best results on both datasets. Specifically, on the Weibo dataset, the gate mode improves the accuracy by 1.3 points compared with the concatenation (Cat) mode. On the Politi dataset, the gate mode improves the accuracy by 2.9 points, compared with both Add and MLP modes. The experimental results suggest the effectiveness of our aggregating function in cross-modal fake news detection.

Table 6

The influence of different feature aggregating functions on the performance. “Add” means to add the contextualized word embeddings, i.e., R_i , with the updated features, i.e., $\tilde{C}_i$ in the image2text fusion module, and to add the contextualized image embeddings, i.e., R_v , with the updated features, i.e., $\tilde{C}_v$ in the text2image fusion module. “GMU” represents using the gated multimodal units (Arevalo, Solorio, Gomez, and Gonzalez, 2017) to aggregate the contextualized embeddings of texts and images. “MLP” indicates that the updated features are nonlinearly activated and then added with the contextualized embeddings. “Cat” denotes concatenating the contextualized embeddings with the updated features.

datasets	aggregating functions	Acc	Fake news Pre	Recall	F1	Real news Pre	Recall	F1
weibo	Add	0.898	0.909	0.893	0.901	0.888	0.904	0.896
	GMU	0.897	0.929	0.866	0.897	0.867	0.930	0.897
	MLP	0.896	0.894	0.905	0.899	0.897	0.886	0.891
	Cat	0.894	0.883	0.915	0.899	0.906	0.870	0.888
	Gate	0.907	0.927	0.889	0.908	0.887	0.925	0.906
politi	Add	0.894	0.800	0.828	0.814	0.932	0.920	0.926
	GMU	0.904	0.788	0.897	0.839	0.958	0.907	0.932
	MLP	0.894	0.846	0.759	0.800	0.910	0.947	0.928
	Cat	0.914	0.833	0.862	0.848	0.946	0.933	0.940
	Gate	0.923	0.862	0.862	0.862	0.947	0.947	0.947



The 60-million-year-old snake fossil was several times larger than the largest living python.

Ya 'an, Sichuan Province, April 20, 2013. A man refused to leave because of an LOL fight. Finally, he was crushed under the room. Fortunately, the network administrator rescued him in time. When rescued, he said "I will live and die with the tower!" His name is Demacia!! This is taking life to play...

Tianjin aunt's neighbor took her 3-year-old son to the supermarket. When she paid the bill, she found that the child was missing and immediately asked the security guard for help. The security staff was very responsible and notified the walkie-talkie to shut down all entrances. The child was eventually found, with his head shaved and his clothes changed. The gangs of human traffickers are so scary that parents should be cautious when taking their children out. Definitely!!

Fig. 3. Qualitative analysis on the Weibo dataset

Notes: The textual information corresponding to Fig. 3(a), (b) and (c) are listed below each figure. (a) The 60-million-year-old snake fossil was several times larger than the largest living python.

(b) Ya 'an, Sichuan Province, April 20, 2013. A man refused to leave because of an LOL fight. Finally, he was crushed under the room. Fortunately, the network administrator rescued him in time. When rescued, he said "I will live and die with the tower!" His name is Demacia!! This is taking life to play...

(c) Tianjin aunt's neighbor took her 3-year-old son to the supermarket. When she paid the bill, she found that the child was missing and immediately asked the security guard for help. The security staff was very responsible and notified the walkie-talkie to shut down all entrances. The child was eventually found, with his head shaved and his clothes changed. The gangs of human traffickers are so scary that parents should be cautious when taking their children out. Definitely!!

4.8. Qualitative analysis

To further verify the proposed BCMF model's effectiveness, we conducted a qualitative analysis by randomly sampling three fake news examples from the experimental results.

As shown in Fig. 3, for the first example, if only the text information of the news is considered, the model cannot accurately predict the authenticity of the news. When the visual features are fused into the model, it can help the model identify the news as fake because the picture has been tampered with. For the second example, the news contains an authentic picture of the earthquake. If the model only leverages visual information, it will identify the news as factual. As the text information is ridiculous, fusing the textual representations into the model can help identify the news as fake. For the third example, it would not be easy to judge the authenticity of the news by the information from a single modality. The concatenation of visual and textual features cannot contribute directly to capturing the mismatch between the information of different modalities. On the contrary, for the proposed BCMF model, the image and text inconsistency has been successfully identified by adding the image2text and text2image fusion modules. The qualitative analysis demonstrates that BCMF can well capture the complementary and matching information between different modalities.

4.9. Visual analysis

In order to demonstrate how the BCMF model distinguishes fake news from real ones, we utilize the t-distributed Stochastic Neighbor Embedding (t-SNE) to reduce the dimension of the obtained integrated textual and visual embeddings. As shown in Fig. 4, the

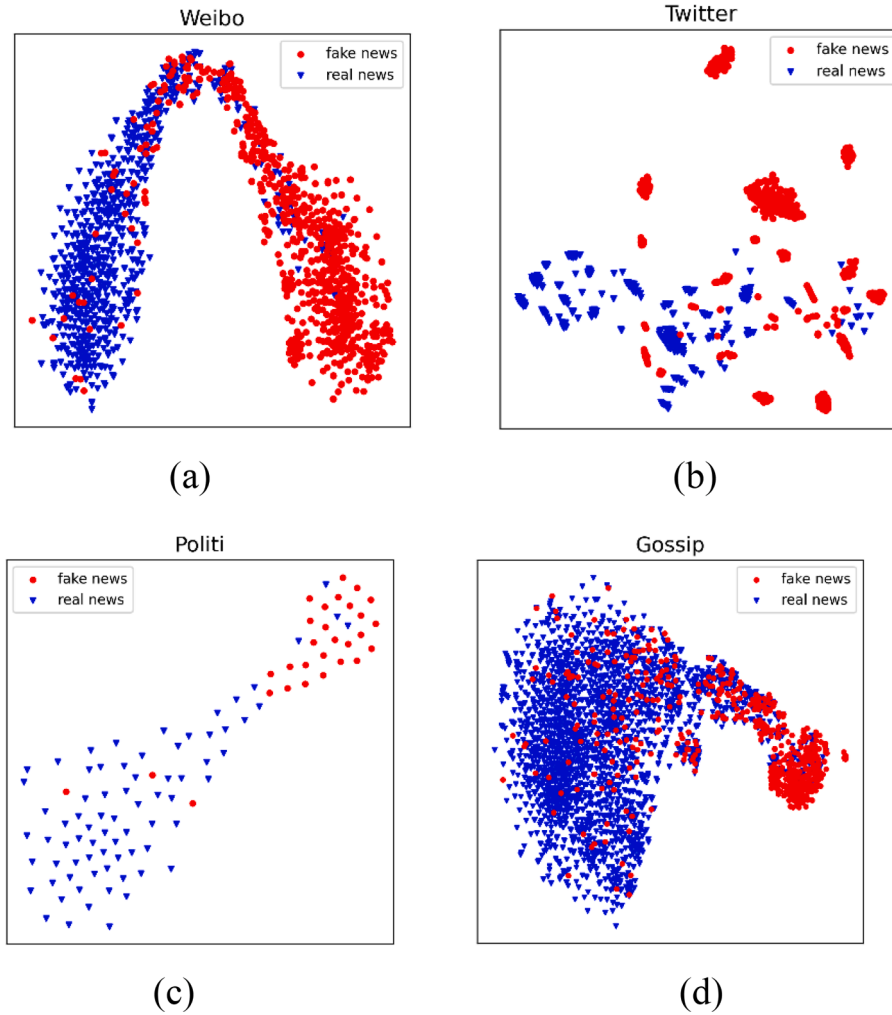


Fig. 4. The visualizations of the multi-modal embeddings on the testing set of Weibo, Twitter, Politi, and Gossip
 Notes: Fig. 4(a), Fig. 4(b), Fig. 4(c), and Fig. 4(d) represent the two-dimensional visualizations on Weibo, Twitter, Politi, and Gossip, respectively. The vertical and horizontal axes for each subgraph represent the two dimensions reduced by the t-SNE method. Specifically, red dots represent fake news samples, and blue triangles represent real news.

subgraph (a), (b), (c), and (d) represent the two-dimensional visualizations on Weibo, Twitter, Politi, and Gossip, respectively.

It can be observed from Fig. 4 that our model achieves good discrimination between fake and real news on all four datasets. The visualizations further validate the effectiveness of the BCMF model. Specifically, concerning the comparison between the two domain-independent datasets, i.e., Weibo and Twitter, it is observed that the model achieves better separations on Weibo than on Twitter. This may be due to the multilingual text in the Twitter dataset and many posts sharing the same picture. Concerning the comparison between the two domain-specific datasets, i.e., Politi and Gossip, it is found that the model has a stronger discriminative ability on the Politi dataset than on the Gossip dataset. This may be due to the highly uneven distribution of samples in the Gossip dataset.

To further validate the effectiveness of the BCMF model on the four datasets, we plot ROC curves, as shown in Fig. 5. We observe from Fig. 5 that the Area Under the Curve (AUC) values are above 0.85 on all the datasets, suggesting our model has achieved good performance. Specifically, comparing the two domain-independent datasets, i.e., Weibo and Twitter, the model achieves a higher AUC value on Weibo than on Twitter. Concerning the comparison between the two domain-specific datasets, i.e., Politi and Gossip, the model has a higher AUC value on the Politi dataset than on the Gossip dataset. The results are consistent with the visualizing results of the multi-modal embeddings on different datasets, as shown in Fig. 4.

4.10. Discussions

4.10.1. Main findings

This study proposes a bidirectional cross-modal fusion model for fake news detection. We investigate three research questions based on this model and have the following findings.

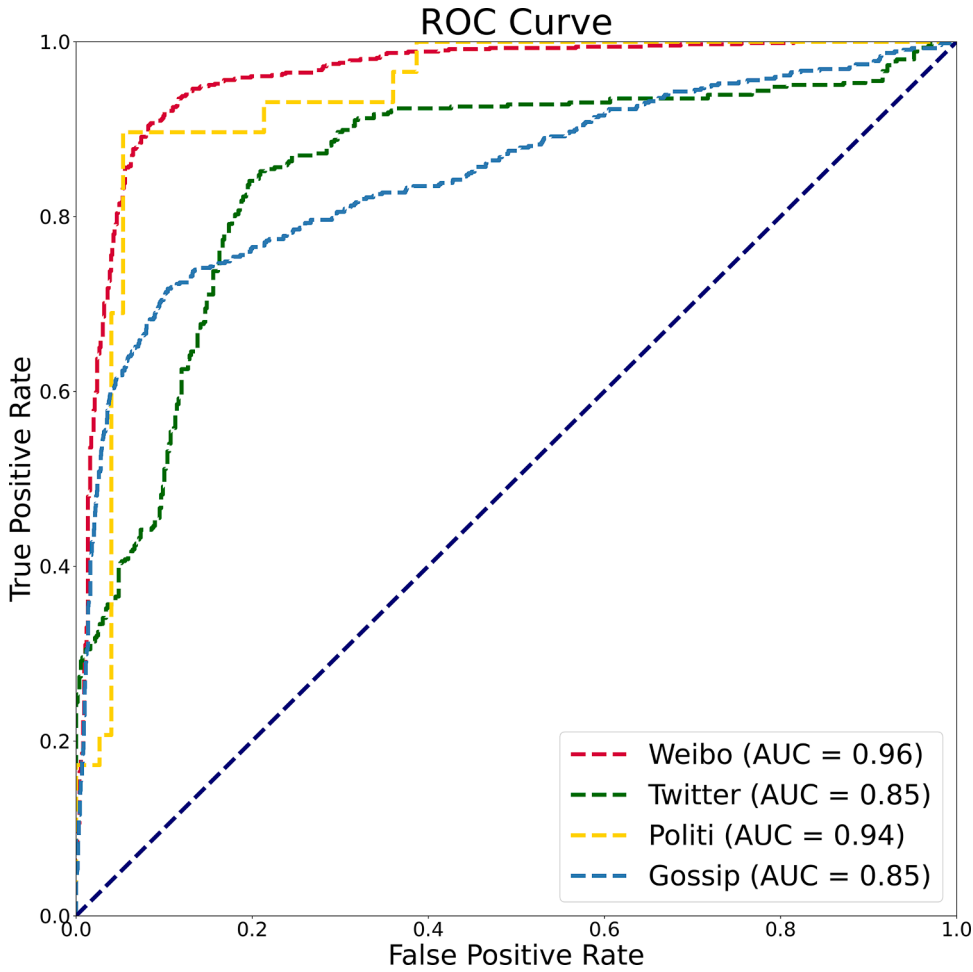


Fig. 5. The ROC curves of the BCMF model on four datasets

Notes: The red, green, yellow, and blue colors represent the ROC curves on Weibo, Twitter, Politi, and Gossip, respectively. The horizontal and vertical axes denote the false positive rate and true positive rate, respectively.

Regarding RQ1, it is found that fake news detection models based on multi-modalities have a significantly better performance than traditional algorithms of single-modality. It confirms the argument of previous works (Khattar et al., 2019; Song, Ning, Zhang, and Wu, 2021) that the utilization of cross-modal information could enhance the performance of fake news detection.

In respect to RQ2, the BCMF model achieves better performance than the state-of-the-art methods. First, compared with methods based on single-modality such as Bi-LSTM (Khattar et al., 2019) and CNN (Singhal et al., 2020), BCMF has significantly better performance due to the supplementary information from different modalities. Second, compared with methods based on modal fusions such as att-RNN (Jin et al., 2017), EANN (Wang et al., 2018), and MVAE (Khattar et al., 2019), the proposed model uses the bi-directional aggregation mechanism to replace the simple concatenation, which leads the improvement of the model effect. Third, compared with the Transformed-based methods such as HMCAN (Qian et al., 2021) and CARMN (Song, Ning, Zhang, and Wu, 2021), BCMF has also achieved better performance. The improvement may be due to the following reasons. (1) In the input embedding module, we employ DeiT to encode the visual information, which alleviates the dependency on the availability of an extensive training set for images; (2) In the aggregation layer, we design two unsymmetric bi-directional fusion modules, i.e., image2text and text2-image, which effectively integrates the semantic representations of different modalities; (3) In the prediction module, we consider the information consistencies between different modalities, which is proved to be more effective.

Regarding RQ3, we have conducted a comparative study to investigate the influence of the model architecture on performance. The experimental results suggest that: (1) adding bi-directional fusions, i.e., the image2text fusion and the text2image fusion, can effectively improve the performance; (2) the number of multi-attention heads does influence the model performance, and there exist an optimum number of multi-attention heads; (3) the aggregating function does influence the performance, and the gating mechanism achieves the best performance.

4.10.2. Theoretical implications

The study has the following theoretical implications. First, it proposes a new perspective by leveraging the multi-modal information for fake news detection. It introduces a mechanism that utilizes bidirectional cross-modal aggregation to enhance the semantic representation of multi-modal information. To the best of our knowledge, the proposed model with empirical evidence is one of the earliest attempts to leverage both contextualized visual embeddings and bi-directional fusions at the same time. As a result, the study sheds light on leveraging cross-modal information to facilitate fake news detection and downstream tasks.

Second, the study reveals the machine factors in misinformation detection. The investigation implies that the improvement of cross-modal fake news detection may be attributed to improving single-modal representation and the integration of different modalities. It also highlights the role of similarities between visual and textual information. The proposed model has achieved the best performance on the Weibo, Wechat, Twitter, Politi, and Gossip datasets. It demonstrates the effectiveness and generalizability of BCMF on both domain-independent and domain-specific datasets.

4.10.3. Practical implications

The study has the following implications for practice. First, it highlights the role of automatic fake news detection in misinformation management. In the real world, it is labor consumptive for an ordinary end-user to differentiate false news from authentic ones. The human ability to judge depends on user characteristics (Zrnec, Pozenel, and Lavbic, 2022). The increase of fake news on social media and other websites, especially after the outbreak of COVID-19, has posed a severe threat to modern society, as the misinformation is causing uncertainty, disturbance, and distrust on a global scale (Barrientos-Baez et al., 2021). The researchers of the information processing and management community have had no choice but to make great efforts to fight against misinformation by providing research support and sharing experiences. This study has affirmed the role of cross-modal information aggregation in the fight against fake news. It suggests that the automatic fusion of information between different modalities can effectively reduce manual labor and be more effective in fake review detection than manual-feature-defined approaches.

Second, the effective detection of fake news is also a fundamental task which substantially influences the performance of the applications based on the news corpus. In recent years, the news corpora has been widely applied in sentiment analysis (Matalon, Magdaci, Almozilino, and Dan, 2021), topic modeling (Rao and Taboada, 2021), event extraction (Repp and Ramampiaro, 2018; Liu et al., 2020), and knowledge base construction (Hürriyetoğlu et al., 2021). Preliminary detection of fake news from the corpora could potentially increase these studies' validity, persuasiveness, dependability, and reliability. Unfortunately, most of current research neglects the process of fake news detection. The proposed model is expected to be implemented to improve the news corpus's quality. The comparative study on the datasets of deleting or keeping fake news, in the case of sentiment analysis, topic modeling, and event extraction, need to be conducted to measure the influence of fake news detection quantitatively. The improvement of these tasks will be further investigated to validate the effect of fake news detection.

Third, this study has laid a foundation for the follow-up study of misinformation identification. For instance, identifying the foci of misinformation in different modalities is a crucial post-task. When we present a process like the one proposed in this study to any authority that needs to use tools to identify foci of misinformation, they may want to know the topics to which that misinformation refers. Regarding the misinformation of textual modality, it is necessary to leverage natural language processing techniques, such as named entity recognition and topic modeling, to identify the critical entities and latent topics. Regarding the misinformation of visual modality, it is necessary to leverage computer vision techniques, such as object detection and image captioning, to identify the visual objects and generate summarization for the image. Based on this, the Spatio-temporal analysis could be carried out to investigate the timeline and geographic distribution of the misinformation. In our future work, we will further investigate the application of fake news identification on these tasks, i.e., multi-modal foci detection and Spatio-temporal analysis, and the influence of the performance of BCMF on them.

5. Conclusions

In this paper, we propose a novel model named Bidirectional Cross-Modal Fusion (BCMF), which comprehensively integrates the textual and visual representations in a bi-directional manner. The proposed model is decomposed into four submodules, i.e., the input embedding, the image2text fusion, the text2image fusion, and the prediction module. After intensive experiments conducted on four real-world datasets, the BCMF model achieves significantly better performance than the state-of-the-art models. Additionally, we find that the bi-directional fusions, the number of multi-attention heads, and the aggregating function impact the performance of the model. The research sheds light on the role of bidirectional cross-modal fusion in leveraging multi-modal information to improve the effect of fake news detection. Our findings have practical implications for enhancing the effect of fake news detection, which could potentially boost the performance of news-corpus-driven applications.

In the follow-up study, we will apply the proposed BCMF model to more domains, especially economy, medical treatment, and law, to further check the influence of domain variations on the performance of the proposed model. In addition, we will further investigate the timeline and geographic distribution of the misinformation promoted by recognizing name entities and detecting topics in a Spatio-temporal analysis.

CRedit authorship contribution statement

Chuanming Yu: Resources, Conceptualization, Methodology, Validation, Writing – original draft, Writing – review & editing, Funding acquisition. **Yinxue Ma:** Methodology, Data curation, Validation, Formal analysis, Writing – original draft. **Lu An:**

Visualization, Methodology, Validation, Writing – review & editing, Funding acquisition. **Gang Li:** Conceptualization, Methodology, Writing – review & editing, Funding acquisition.

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